Reflection AI Beam

Research Baseline for Building a Beam-Class Open MoE Language Model

Published details · architecture · benchmarks vs. Anthropic, OpenAI, Google, Meta and open-weight peers · KPI targets · recommendations · download links

Prepared for: Yossi Avni

Date: 14 October 2026

Information cut-off: public sources available up to 13–14 October 2026 (nine days after Beam's announcement)

Status of Beam weights at cut-off: announced, not yet published (Apache 2.0 promised for October 2026)

Reading guide: every Beam number in this document is vendor-reported unless marked otherwise. Items marked "estimate" or "" are analyst or author inferences, not Reflection disclosures. Re-check Section 11 links and the Beam technical report once released; several architecture fields are expected to be filled in by it.*

Contents

Right-click the table of contents in Word and choose “Update Field” to refresh page numbers.

1. Executive summary

Reflection AI announced Beam on 5 October 2026: its first open-weight model, a sparse Mixture-of-Experts (MoE) transformer with 501 billion total parameters and about 23 billion active per token ("501B-A23B"). It is text-only, aimed at coding, reasoning and agentic workloads, was pretrained from scratch on 23.8 trillion tokens, extended to a 1-million-token context, and then post-trained with an unusually large reinforcement-learning (RL) run on 10,500 NVIDIA GB300 GPUs for about four weeks.

Its headline claim is efficiency rather than raw leadership: Reflection says Beam matches Z.ai's GLM-5.2 on advanced reasoning with 3–4× less inference compute. On Reflection's own table Beam trails Kimi K3 and Qwen 3.8 Max on most rows, is roughly at parity with GLM-5.2, and sits clearly behind the closed frontier (Claude Fable 5.1, GPT-6-series, Gemini 3.x) on the hardest benchmarks such as Humanity's Last Exam.

Key takeaways for a team building a Beam-equivalent model

- **The target is achievable at mid-scale.** Beam is the smallest model in its competitive set (Figure 3). A ~500B-A20–25B MoE trained on 20–25T curated tokens plus heavy agentic RL is the right reference design.
- **RL compute is the differentiator.** Beam's RL phase used ~1.7× the GPUs of pretraining for a similar duration, across ~1M environments and >100M rollouts. Budget for post-training compute ≥ pretraining compute.
- **Data quality over quantity.** Reported curation discarded ~95% of raw web tokens. A data engine (filtering, synthetic code/math, executable environments) is the longest-lead item in the plan.
- **Architecture is mainstream 2026 practice.** Interleaved sliding-window/global attention, fine-grained experts, auxiliary-loss-free balancing, SandwichNorm, gated attention and FP32 residuals are all published techniques; none is proprietary.
- **Budget estimate:** ~10M GB300-hours in total (≈2–3M pretraining + ≈7M RL), on the order of US$30–60M of compute at market rates, plus data and staff (Section 9).

Terminology: the request mentioned "PKI"; this document interprets it as KPI (Key Performance Indicators) — the benchmark and efficiency targets a new model must reach. Section 8 defines them.

2. Reflection AI: company background

Item	Detail
Founded	2024, by former Google DeepMind researchers
Founders	Misha Laskin (CEO); Ioannis Antonoglou (CTO, co-creator of AlphaGo)
Mission	An American open frontier lab: open-weight models competitive with leading Chinese open models (DeepSeek, Qwen, Kimi, GLM)
Funding	US$2B round Oct 2025 led by NVIDIA (~US$800M), at US$8B valuation; reported talks in Mar 2026 for US$2.5B at ~US$25B pre-money; one aggregator lists US$27.5B (May 2026, unconfirmed)
First product	Asimov — a code-research agent (waitlist in 2025–26)
First model	Beam, announced 5 Oct 2026; weights promised under Apache 2.0 in October 2026
Recent news	Reports (Oct 2026) of early talks in which NVIDIA could acquire, invest further in, or license technology from Reflection — a reason to mirror weights promptly once released

Do not confuse with "Reflection Llama-3.1 70B" (2024, by Matt Shumer / HyperWrite) — an unrelated fine-tune with a similar name.

3. Beam: everything published so far

Attribute	Value	Confidence / source
Model type	Sparse Mixture-of-Experts decoder-only transformer	Reflection launch post
Total parameters	501B	Reflection
Active parameters / token	~23B (≈4.6% of total)	Reflection
Layers	52	Secondary coverage
Attention	Interleaved local (sliding-window) and global attention	Reflection; 3:1 ratio is an analyst reading
Attention gating	Elementwise gating on attention output	Reflection
Normalisation	SandwichNorm (pre- and post-sublayer norm)	Reflection
Residual stream	Depth-based residual scaling; FP32 residual accumulation	Reflection
Experts	Fine-grained routed experts, learned top-k router; counts not disclosed	Reflection
Load balancing	Auxiliary-loss-free (expert-bias) balancing with cosine decay of bias updates; busiest expert ≈1.04× mean load	Reflection
Modality	Text only	DataCamp, TrendingTopics
Pretraining data	23.8T tokens; ~95% of raw web tokens discarded in curation	Reflection; curation figure from MarkTechPost
Pretraining compute	6,144 NVIDIA GB300 GPUs, < 4 weeks	Reflection / coverage
Context window	1M tokens after mid-training; 262,144 tokens in beta API	MarkTechPost; Kingy AI
RL phase	10,500 GB300 GPUs, ~4 weeks, >100M rollouts, ~1M coding/agentic/STEM environments, ~1.3B sandboxes, 256K max rollout context	Reflection; MarkTechPost
Efficiency claim	Matches GLM-5.2 on reasoning at 3–4× lower inference compute (excludes prefill and serving overhead)	Reflection; RuntimeWire caveat
License	Apache 2.0 (promised)	Reflection
Availability (14 Oct 2026)	Waitlisted beta API; weights, model card and technical report not yet published	OrcaRouter, OpenModelWeights
Pricing	No public rate card; not yet on Artificial Analysis index	eesel.ai, OrcaRouter
Not disclosed	Tokenizer & vocabulary, hidden size, head counts, expert count / top-k / shared experts, optimizer, MTP, precision recipe	—

4. Architecture deep-dive

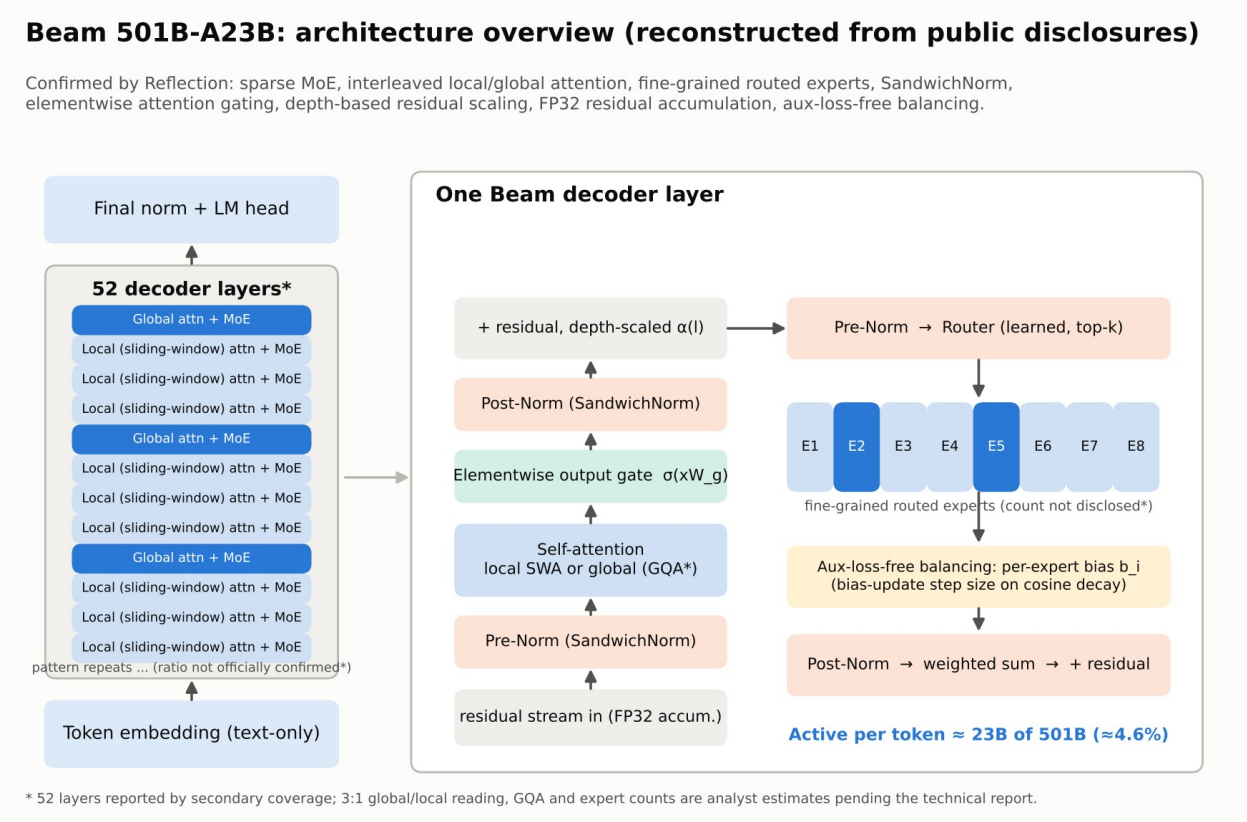


Figure 1 — Beam architecture reconstructed from public disclosures. Starred items await the technical report.

4.1 Sparse MoE backbone

Every layer replaces the dense feed-forward block with a bank of many small (“fine-grained”) experts. A learned router scores all experts for each token and dispatches it to the top-k. Only ~23B of 501B parameters are touched per token, so per-token FLOPs resemble a ~23B dense model while capacity resembles a ~500B model. Fine-grained experts (DeepSeekMoE lineage) give the router more combinations and improve specialisation.

4.2 Auxiliary-loss-free load balancing with cosine-decayed bias updates

Instead of adding a balancing loss that fights the language-modelling objective, each expert carries a bias b_i that is added to its routing score only for top-k selection. After each step the bias is nudged up for under-loaded experts and down for over-loaded ones. Reflection’s contribution is to decay the bias-update step on a cosine schedule so routing stabilises late in training. Reported result: the busiest expert averages only 1.04× the mean load — near-perfect utilisation, which matters for expert-parallel throughput.

4.3 Interleaved local / global attention

Most layers use sliding-window (local) attention whose KV cache is bounded by the window; periodic global layers attend over the full sequence. This cuts KV-cache memory and long-context FLOPs roughly in proportion to the share of local layers, which is how a 1M-token context becomes affordable. The same pattern appears in Gemma 3, MiMo-V2.6 and other 2026 models.

4.4 Stability components

- **SandwichNorm:** normalisation both before and after each sub-layer, bounding activation growth in deep, sparse networks.
- **Elementwise attention gating:** a learned sigmoid gate on the attention output; known to suppress attention sinks and loss spikes and to help long-context extrapolation.
- **Depth-based residual scaling:** scaling residual branch contributions by layer depth for stable signal propagation.
- **FP32 residual accumulation:** the residual stream is accumulated in FP32 even when matmuls run in low precision (BF16/FP8), avoiding drift over 52 layers.

5. Training recipe

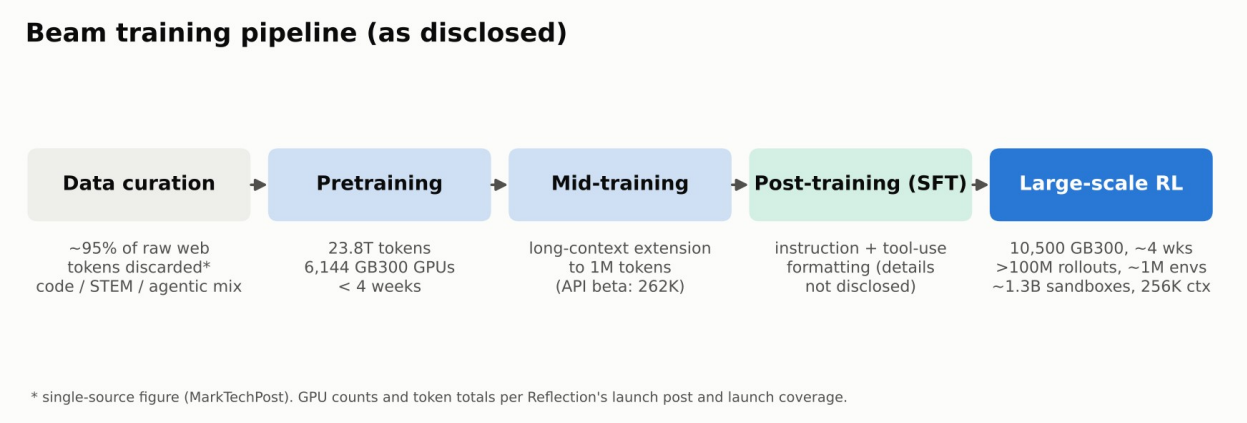


Figure 2 — Beam's disclosed training pipeline.

Phase	Disclosed facts	What it implies for a replication
Data curation	Aggressive filtering (~95% of raw web discarded)	Invest in classifiers, dedup, decontamination and synthetic code/math before training
Pretraining	23.8T tokens on 6,144 GB300 GPUs in < 4 weeks	≈ 3.3×10^24 training FLOPs (6·N_active·D); needs ≈ 20–35% MoE MFU
Mid-training	Context extension to 1M tokens	Staged RoPE/window extension with long-document and repository-level data
SFT	Not detailed	Tool-use, multi-turn agent traces and reasoning formats seed the RL policy
RL	10,500 GB300, ~4 weeks, >100M rollouts, ~1M environments, ~1.3B sandboxes	Asynchronous RL infrastructure, sandbox orchestration and verifiable rewards are core engineering

Observation: In GPU-hours Beam's RL phase (≈7.1M GB300-hours) was roughly 2–3× its pretraining phase. This inverts the classic ratio and reflects the 2026 consensus that agentic capability is mostly bought in post-training.

6. Performance vs. leading models

6.1 Model scale in context

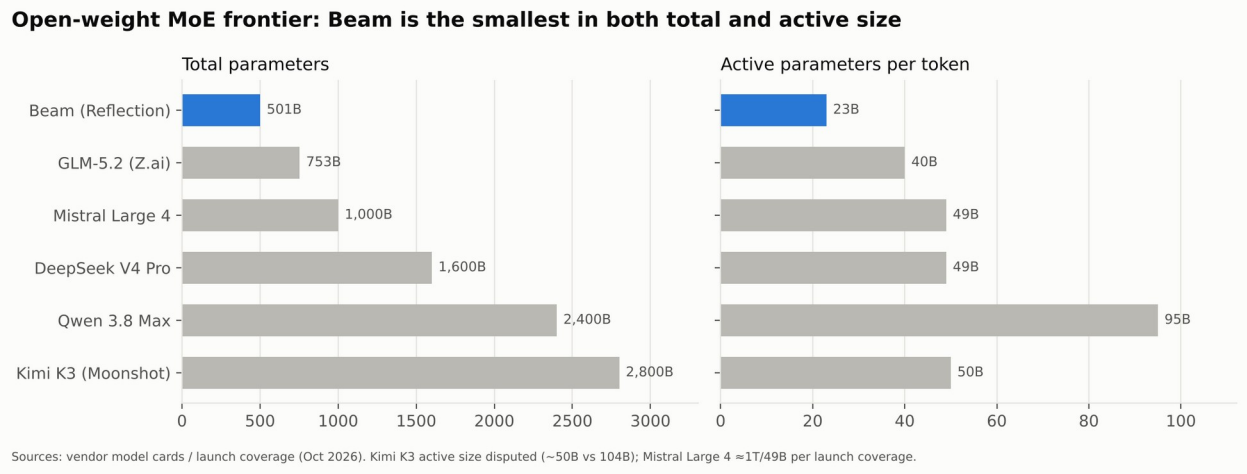


Figure 3 — Total and active parameters of Beam and its open-weight competitors.

Model (developer)	Total params	Active params	Open weights?
Beam (Reflection AI)	501B	~23B	Promised, Apache 2.0
GLM-5.2 (Z.ai / Zhipu)	~753B	~40B	Yes
Mistral Large 4 (Mistral AI)	~1T	~49B	Per launch coverage
DeepSeek V4 Pro (DeepSeek)	~1.6T	~49B	Yes
Qwen 3.8 Max (Alibaba)	~2.4T	~95B	Base checkpoint; hosted Max adds features
Kimi K3 (Moonshot AI)	~2.8T	~50B (disputed; some report 104B)	Yes
Claude Fable 5.1 / Opus 5.5 (Anthropic)	Undisclosed	Undisclosed	No (API only)
GPT-6 Astra / GPT-6.1 Sol (OpenAI)	Undisclosed	Undisclosed	No (API only)
Gemini 3.1 Pro / 3.8 Flash (Google)	Undisclosed	Undisclosed	No (API only)
Llama 4 Maverick (Meta)	~400B	~17B	Yes (Llama license)

6.2 Beam vs. open-weight peers (Reflection's launch table)

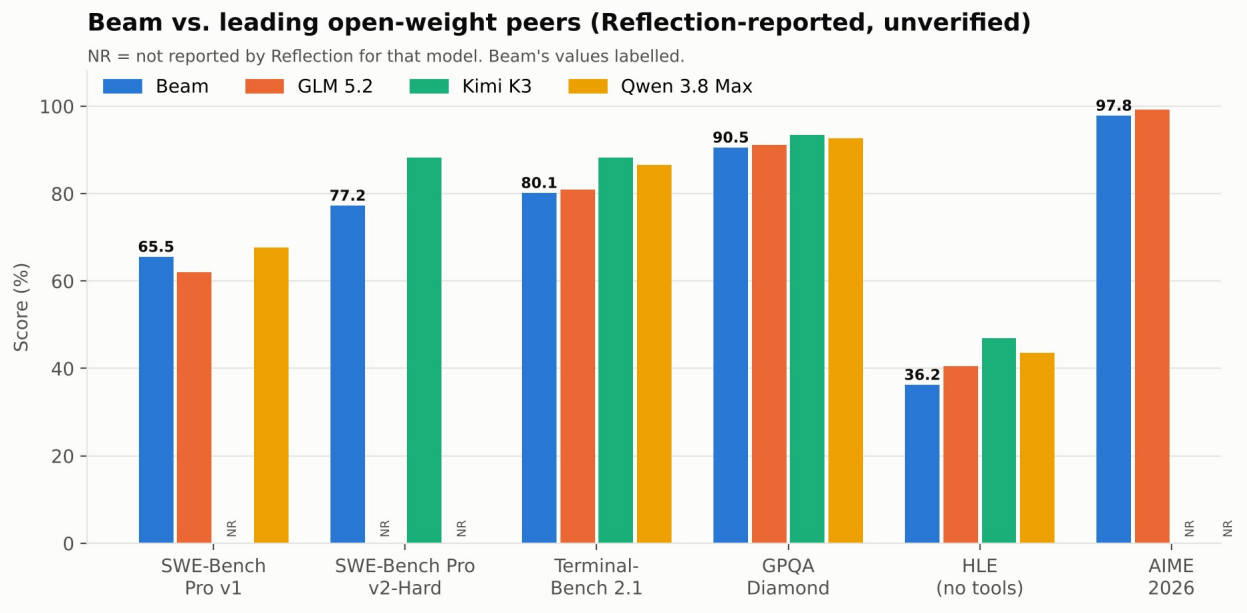


Figure 4 — Reflection-reported scores; NR = not reported.

Benchmark	Beam	GLM 5.2	Kimi K3	Qwen 3.8 Max
SWE-Bench Pro v1	65.5	62.1	NR	67.7
SWE-Bench Pro v2-Hard	77.2	NR	88.2	NR
SWE-Bench Verified	80.9	NR	NR	NR
Terminal-Bench 2.1	80.1	81.0	88.3	86.6
GPQA Diamond	90.5	91.2	93.5	92.6
HLE (no tools)	36.2	40.5	46.9	43.6
AIME 2026	97.8	99.2	NR	NR

Reading: Beam beats GLM-5.2 on SWE-Bench Pro v1 and is within ~1 point on Terminal-Bench and GPQA, but trails it by 4 points on HLE. Kimi K3 and Qwen 3.8 Max lead every row on which they report. Beam's value proposition is therefore per-token efficiency (Figure 6), not absolute leadership.

6.3 Beam vs. the closed frontier (Anthropic, OpenAI, Google) and Meta

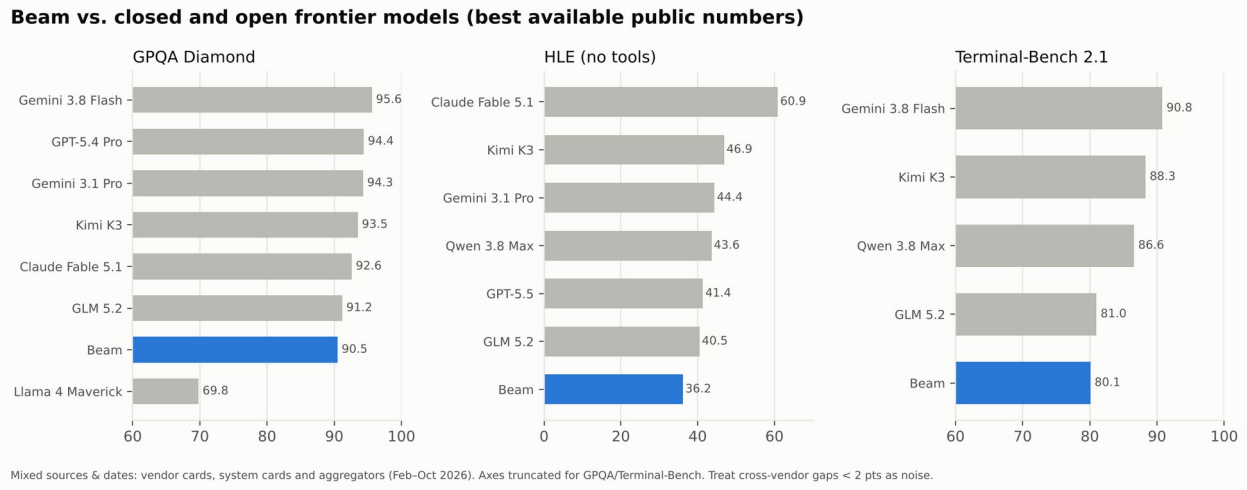


Figure 5 — Best available public numbers; versions and dates differ across vendors.

Model	GPQA Diamond	HLE (no tools)	Terminal-Bench 2.1	Notes
Beam (Reflection)	90.5	36.2	80.1	Vendor-reported, Oct 2026
Claude Fable 5.1 (Anthropic)	92.6	60.9 (65.0 w/ tools)	—	SWE-bench Pro 81.2; Terminal-Bench 4.0 55.8 (Sep 2026 system card)
Claude Opus 5.5 (Anthropic)	—	—	—	Comparable published scores not found; trails GPT-6.1 Sol by 2.2 pts on AutomationBench per one source
GPT-6.1 Sol (OpenAI)	—	—	—	Terminal-Bench Science 57.0; OSWorld 2.0 71.4; DeepSWE 71.9–75.2
GPT-5.4 Pro / GPT-5.5 (OpenAI)	94.4 (5.4 Pro)	~41.4 (5.5)	—	Earlier 2026 generation
Gemini 3.8 Flash (Google)	95.6	54.9 (HLE-Verified)	90.8 (89.4 alt.)	Sep 2026; SWE-Bench Pro 61.6
Gemini 3.1 Pro (Google)	94.3	44.4	68.5 (v2.0)	Feb 2026
Llama 4 Maverick (Meta)	69.8	—	—	Apr 2025; latest open Llama; Meta's new Muse line is closed
Groq	n/a	n/a	n/a	Groq builds LPU inference hardware and hosts others' open models; it does not train frontier LLMs

How to read the frontier comparison

- Closed-model vendors increasingly publish newer benchmark versions (Terminal-Bench 4.0, SWE-bench Pro, OSWorld 2.0) while open-model vendors publish Terminal-Bench 2.1 and SWE-Bench Pro v1/v2, so direct like-for-like numbers are sparse. Gaps under ~2 points are within harness noise.

- On the least-saturated benchmark (HLE, no tools) Beam's 36.2 is ~25 points behind Claude Fable 5.1 and ~8 points behind Gemini 3.1 Pro — the clearest measure of the frontier gap.
- GPQA Diamond is near saturation (90–96% for all frontier and top open models); it no longer separates models well.
- Groq is relevant as a deployment target (very high tokens/s for open MoE models) rather than as a competitor. NVIDIA licensed Groq's LPU technology in Dec 2025 (~US$20B).

6.4 Efficiency: reasoning per active parameter

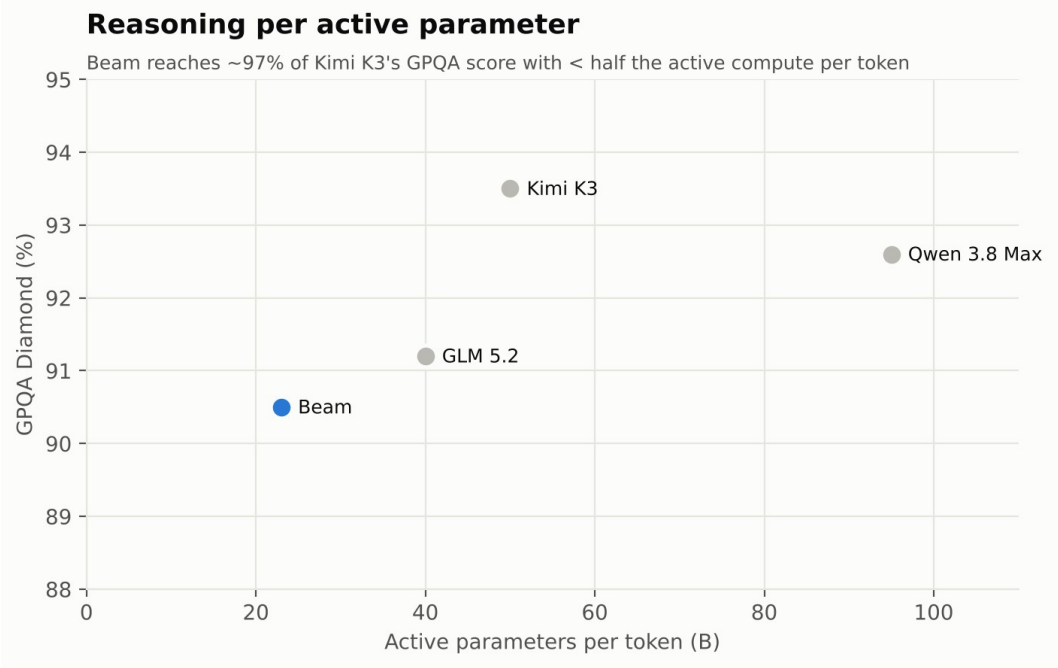


Figure 6 — GPQA Diamond vs. active parameters per token (open-weight models).

Beam reaches ~97% of Kimi K3's GPQA score with under half of its active parameters, and ~98% of Qwen 3.8 Max's with under a quarter. For serving cost, active parameters drive decode FLOPs while total parameters drive memory footprint; at 501B total, Beam fits on a single 8×GB300 node in FP8 (≈0.5 TB weights) — a practical advantage over 1.6–2.8T-parameter peers.

7. What is still unknown (open questions)

Unknown	Why it matters for replication	Expected source
Expert count, top-k, shared experts, expert width	Determines active/total ratio, routing granularity and expert-parallel layout	Technical report / config.json
Hidden size, heads, KV heads, window size, global ratio	Determines KV-cache size and long-context cost	config.json
Tokenizer & vocabulary	Affects code/multilingual efficiency and compatibility	Tokenizer files
Optimizer (AdamW vs. Muon) and LR schedule	Muon has shown ~2× token efficiency at scale	Technical report
Precision (BF16 / FP8 / NVFP4)	Drives GB300 throughput and stability	Technical report
Data mixture proportions	Biggest driver of coding/agentic quality	Technical report (partial at best)
RL algorithm & reward design	GRPO/PPO variant, async off-policy correction, reward models vs. verifiers	Technical report
Independent benchmark results	All scores are vendor-reported	Artificial Analysis, LMArena, Vals, SWE-bench leaderboards

8. KPI targets for a Beam-equivalent model

A new model is “Beam-equivalent” if it meets the following KPIs on an identical evaluation harness. Run Beam itself through the same harness once weights ship — that is the only fair baseline.

KPI	Beam reference	Minimum target	Stretch target
SWE-Bench Pro v1	65.5	≥ 65	≥ 68 (Qwen 3.8 Max)
SWE-Bench Verified	80.9	≥ 80	≥ 83
Terminal-Bench 2.1	80.1	≥ 80	≥ 86
GPQA Diamond	90.5	≥ 90	≥ 92
HLE (no tools)	36.2	≥ 36	≥ 42
AIME 2026	97.8	≥ 97	≥ 99
Active parameters / token	~23B	≤ 25B	≤ 20B
Total parameters	501B	≤ 550B	≤ 400B
Context window (effective)	1M (262K served)	≥ 256K with RULER ≥ 85%	1M
Decode throughput, 8×GB300 FP8	Not published	Measure Beam; match it	+20%
Inference compute vs. GLM-5.2 at equal quality	3–4× lower	≥ 3× lower	≥ 4× lower
Expert load imbalance (max / mean)	1.04×	≤ 1.10×	≤ 1.05×
License	Apache 2.0	Permissive open weights	Apache 2.0 + open data card

9. Proposed reference design and compute budget

9.1 Starting-point configuration (to be validated by ablation)

Component	Proposed choice	Rationale
Size	~480–520B total, 20–24B active, ~52–60 layers	Matches Beam; fits one 8-GPU node in FP8
Experts	256 routed fine-grained experts + 1 shared, top-8	DeepSeek-V3/V4-style granularity; ablate 128/256/384
Routing	Aux-loss-free bias balancing with cosine-decayed update rate; sequence-level balance check	Reproduces Beam's 1.04× load
Attention	3:1 sliding-window (4–8K window) : global, GQA or MLA on global layers, gated output	Bounded KV cache, 1M-context capable
Norm / residual	SandwichNorm (RMSNorm), depth-scaled residuals, FP32 residual accumulation, QK-norm	Stability at low precision
Position encoding	RoPE on global layers with YaRN-style extension; NoPE/partial RoPE as ablation	Long-context extrapolation
Prediction head	Multi-token prediction (1–2 extra heads)	Denser training signal + speculative decoding
Tokenizer	~160–200K BPE vocabulary, code- and multilingual-aware	Compression on code and non-English text
Optimizer	Muon for matrix params (per-expert), AdamW for embeddings/norms	~2× token efficiency in published studies
Precision	FP8 (block-scaled) training, ablate NVFP4 on GB300	Throughput on Blackwell-class GPUs

9.2 Compute and cost estimate

Phase	Estimate	Assumptions
Pretraining	≈ 3.3×10^24 FLOPs → ≈ 2–3M GB300-hours	6 × 23B active × 23.8T tokens; 20–30% MFU on ~2.5 PFLOP/s BF16-dense (higher with FP8)
Mid-training (long context)	≈ 0.2–0.4M GB300-hours	~1T tokens at growing sequence length
RL	≈ 7M GB300-hours	10,500 GPUs × 28 days, as disclosed for Beam
Ablations & scaling pilots	≈ 1–2M GPU-hours	1–7B and ~60B-A3B models
Total	≈ 10–12M GPU-hours	≈ US$30–60M at US$3–5 per GPU-hour, excluding staff and data

These are author estimates derived from Beam's disclosed GPU counts and durations, not Reflection figures.

9.3 Program roadmap

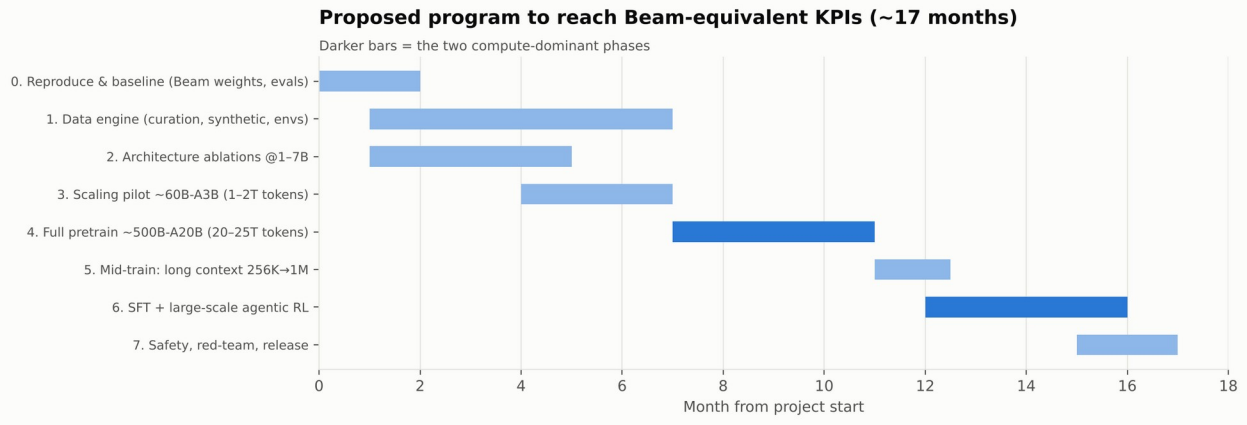


Figure 7 — Proposed ~17-month program to reach Beam-equivalent KPIs.

10. Recommendations from current research

10.1 Architecture

1. **Adopt fine-grained MoE with aux-loss-free balancing from day one.** It is now standard (DeepSeek-V3/V4, Beam). Add Reflection's cosine decay on bias updates and monitor max/mean expert load per layer.
2. **Use hybrid attention for long context.** Interleave sliding-window and global layers (Beam, MiMo-V2.6); evaluate compressed/sparse global attention as in DeepSeek-V4 for 1M-token contexts, and dynamic sparse attention for decode.
3. **Gate attention outputs and use SandwichNorm + QK-norm.** Cheap insurance against loss spikes during multi-trillion-token runs.
4. **Add multi-token prediction.** Improves sample efficiency and gives a built-in speculative decoder (used by DeepSeek-V3/V4, MiMo-V2.6).

10.2 Optimisation and precision

5. **Evaluate Muon.** Moonlight reported ~2× compute efficiency vs AdamW; DeepSeek-V4 trains with Muon and orthogonalises each expert separately; 2026 studies find it scales better with batch size.
6. **Train in FP8, pilot NVFP4.** Blackwell/GB300 FP4 throughput is roughly double FP8; keep FP32 master weights and residual accumulation as Beam does.
7. **Use WSD (warmup-stable-decay) learning-rate schedules** so that intermediate checkpoints can be annealed for evaluation and mid-training branches.

10.3 Data

8. **Build the data engine first.** Model-based quality classifiers, aggressive dedup, benchmark decontamination, and synthetic code/math with execution-verified answers. Beam's ~95% discard rate signals how strict this must be.
9. **Collect repository-level and long-document data** for mid-training; include agent traces (terminal sessions, tool calls) for SFT.

10.4 Post-training and agentic RL

10. **Plan RL compute ≥ pretraining compute.** Beam's 10,500-GPU RL run and MiMo-V2.6's RL-for-self-improvement framing both point the same way.
11. **Invest in environments, not just prompts.** Target 10^5–10^6 verifiable environments (unit-test-graded repos, terminal tasks, math/STEM checkers, web/tool tasks) with a sandbox fleet that scales to billions of executions.
12. **Use asynchronous, off-policy-corrected RL** (GRPO-family objectives with importance-ratio clipping) to keep inference and training GPUs saturated; support 128K–256K rollout context.

10.5 Evaluation

13. **Evaluate on unsaturated, contamination-resistant suites:** SWE-bench Pro (OpenAI no longer uses SWE-bench Verified for frontier coding), Terminal-Bench 2.x/4.0, HLE, AIME of the current year, RULER/long-context, and agentic suites (OSWorld, τ-bench).
14. **Report with confidence intervals and fixed harnesses;** re-run competitors yourself. Vendor-to-vendor gaps of 1–2 points are noise.

10.6 Strategy

15. **Fine-tune before you pre-train.** Once Beam (Apache 2.0) ships, a Beam- or GLM-based fine-tune is the fastest way to validate your data, RL stack and evaluation harness before committing ~US$50M to pretraining from scratch.
16. **Differentiate on one axis.** Beam's angle is efficiency per active parameter; pick yours (e.g., domain specialisation, multimodality, on-device distillation, or a sub-10B-active model).
17. **Mirror open checkpoints immediately on release** — Apache 2.0 grants are irrevocable, but hosting can change if ownership does.

11. Download links and access

Beam status (14 Oct 2026): weights are not yet downloadable. Reflection has promised Apache 2.0 weights, a model card and a technical report in October 2026; early access is via a waitlisted beta API.

Model / resource	Where	Status
Reflection Beam — announcement	reflection.ai/blog/introducing-beam	Live
Reflection Beam — weights	huggingface.co (search “reflection”)	Pending (promised Oct 2026)
DeepSeek V4 family	huggingface.co/deepseek-ai	Open weights
Kimi K3 (Moonshot)	huggingface.co/moonshotai	Open weights
GLM-5.x (Z.ai)	huggingface.co/zai-org	Open weights
Qwen 3.x (Alibaba)	huggingface.co/Qwen	Open weights
Mistral (Mistral AI)	huggingface.co/mistralai	Open weights (check license per model)
Llama 4 (Meta)	huggingface.co/meta-llama	Gated, Llama license
gpt-oss (OpenAI open-weight)	huggingface.co/openai	Open weights
Claude (Anthropic)	console.anthropic.com	API only
GPT-6 family (OpenAI)	platform.openai.com	API only
Gemini (Google)	aistudio.google.com	API only
GroqCloud (fast inference of open models)	console.groq.com	API (hosted open models)

Clickable links:

- Reflection — Introducing Beam
- Hugging Face search: Reflection models
- DeepSeek on Hugging Face
- Moonshot AI (Kimi) on Hugging Face
- Z.ai (GLM) on Hugging Face
- Qwen on Hugging Face
- Mistral AI on Hugging Face
- Meta Llama on Hugging Face
- OpenAI on Hugging Face
- Anthropic Console
- OpenAI Platform
- Google AI Studio
- GroqCloud Console

Exact Beam repository IDs could not be verified before release; use the Hugging Face search link and confirm the publisher is Reflection’s verified organisation.

12. Sources

[1] Reflection — Introducing Beam (launch post)
[2] Reflection on X — Beam announcement
[3] TechCrunch — Reflection debuts Beam
[4] SiliconANGLE — Beam with 501B parameters
[5] Fortune — Reflection AI unveils Beam
[6] MarkTechPost — Beam 501B open-weight MoE
[7] DataCamp — Beam explained
[8] MindStudio — Beam explained
[9] Latent Space AINews — Reflection Beam 501B-A23B
[10] Kingy AI — Specs, benchmarks & API limits
[11] Munder Difflin — Beam explained
[12] RuntimeWire — Beam benchmark scores
[13] OrcaRouter — Weights not out
[14] eesel.ai — Specs, access and price
[15] OpenModelWeights — Beam preview
[16] Wikipedia — Reflection AI
[17] Anthropic — Claude Fable 5.1 & Mythos 5.1 System Card
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[19] Vellum — GPT-6.1 Sol benchmarks
[20] BenchLM — GPT-6.1 Sol
[21] DataCamp — Gemini 3.8 Flash
[22] OpenRouter — GPQA Diamond leaderboard
[23] Digital Applied — Gemini 3.1 Pro benchmarks
[24] ComputingForGeeks — Open-source LLM comparison 2026
[25] DeepInfra — Kimi K3 vs DeepSeek V4 Pro vs GLM-5.2
[26] Codersera — Llama 4 guide 2026
[27] Voiceflow — Groq in 2026
[28] DeepSeek-V4 technical report (arXiv)
[29] MiMo-V2.6: Scaling RL Towards Self-Improvement (arXiv)
[30] Muon is Scalable for LLM Training (arXiv)
[31] SOAP, Muon, and Beyond (arXiv)
[32] SWE-Bench Pro Verified (arXiv)

Appendix A. Glossary

Term	Meaning
MoE	Mixture-of-Experts: layers with many expert sub-networks, only a few active per token
501B-A23B	501B total parameters, 23B active per token
Aux-loss-free balancing	Load balancing via per-expert routing biases instead of an extra loss term
SWA	Sliding-window attention: each token attends only to a recent window
GQA / MLA	Grouped-query / multi-head latent attention: ways to shrink the KV cache
SandwichNorm	Normalisation before and after each sub-layer
MTP	Multi-token prediction: extra heads predicting tokens further ahead
Muon	Optimizer that orthogonalises momentum updates for matrix parameters
MFU	Model FLOPs Utilisation: achieved / peak hardware FLOPs
GPQA Diamond	Graduate-level science multiple-choice benchmark
HLE	Humanity's Last Exam: expert-written questions at the frontier of knowledge
SWE-Bench Pro / Verified	Real GitHub issue-fixing benchmarks; Pro is harder and less contaminated
Terminal-Bench	Agentic tasks performed in a real terminal
AIME	American Invitational Mathematics Examination problems